Exercise 2 : SQL Aggregate functions & Operators

01. SELECT DISTINCT department
FROM students;

Department
IT
HR
IT
Finance
HR

2. SELECT department
AVG(age) AS avg-age
FROM students
GROUP BY department;

Department	Avg-age
IT	20,5
HR	22
Finance	23

3. SELECT department, COUNT(*) AS student-count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;

Department	student-count
IT	2
HR	2

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4. SELECT Student-id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;

Student-id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

5. SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT', 'HR')
AND age > 21

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

The Courses Table

6. SELECT department, SUM (credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM (credit) > 5

department	total_credits
IT	11

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7. SELECT course-id, course-name, department, Credits
FROM courses
WHER Credits <> 4;

Course-id	Course-name	department	Credits
101	SQL Basics	IT	3
104	Excel	finance	2
105	statistics	HR	3

8. SELECT course-id, course-name, Credits
FROM Courses
ORDER BY credits DESC
LIMIT 3,

Course-id	Course-name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

The Enrollments table

9. SELECT
MAX (grade) AS max-grade,
MIN (grade) AS min-grade,
AVG (grade) AS Avg-grade
FROM ourollments;

max-grade	min-grade	avg-grade
90	78	84.6

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10. SELECT course_id,
COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;

course_id	enrollment_count
C01	1
C02	1
C03	1
C04	1
C05	1

The salaries table

11. SELECT department,
SUM(salary) AS total_salary,
SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;

department	total_salary	total_bonus
IT	120 000	10 500
HR	109 000	7500
finance	70 000	6 000

12. SELECT department,
AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000

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department	avg_salary
IT	61 000
finance	70 000

13. SELECT employee_id, name, salary, bonus, salary, ~~bonus~~ ~~sta~~ salary + bonus AS total_compensation
FROM ~~salary~~ salaries
WHERE salary + bonus > 60 000;

employee_id	name	salary	bonus	total_compensation
1	Tom	60 000	5000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 600	5500	67 500

The project table

14. SELECT department,
SUM (budget) AS total_budget,
AVG (budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70 000;

department	total_budget	avg_budget
IT	270 000	135 000
finance	80 000	80 000

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15. SELECT project-id, project_name, department, budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
AND department <> 'Marketing';

project-id	project-name	department	Budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000

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